The stylized model

Demography and timing

Overlapping generations. A household lives three periods — **child**, **young**, **old** — with value functions W_0, W_1 and W_2. Each cohort has unit mass. The child period is spent in the parent's household and carries no choices: the child consumes what the parent's allocation gives it. A household has one child *while young*; the child leaves at the end of that period and becomes young itself, so this period's children are next period's young and the cohort entering is unit mass forever. The equivalence scale is therefore ξ while young and 1 when old: the old carry no child, in the scale or in the objective.

The disorder is realised at the *transition* from young to old, and survival over that transition depends on it. It cannot be realised inside the final period, where death costs nothing because nothing follows it.

Period 1

State. Permanent income y. An employment draw and, if income-eligible, a voucher draw give realised income and a rent multiplier (inc, r). Realised income is floored at $\underline{y}$. A voucher is means tested at y_V, arrives with probability π_V, and cuts the tenant's rent to κ_r of the market rent.

Choices. Drug purchase $v \geq 0$ at price p_v, housing $h \in \{h_1, h_2\}$ — the street, which is free and supplies h_1, or the one rental unit, which supplies h_2 at rent $p_r h_2$ — and saving $a' \geq 0$, subject to

$$c = \text{inc} - p_v v - \text{rent}(h, r) - a' > 0.$$

Flow utility. With $\varphi_h = \gamma(1-\theta)$, $\varphi_c = (1-\gamma)(1-\theta)$, $\varphi_q = \theta$ and $u(x) = x^{1-\sigma}/(1-\sigma)$,

$$X_P = \left(\tfrac{c}{\xi}\right)^{\varphi_c} \left(\tfrac{h}{\xi}\right)^{\varphi_h} (\bar{b} + v)^{\varphi_q}, \qquad X_K = \left(\tfrac{c}{\xi}\right)^{\varphi_c} \left(\tfrac{h}{\xi}\right)^{\varphi_h} \bar{b}^{\,\varphi_q},$$

$$\text{flow} = u(X_P) + \omega_c(\xi - 1)\, u(X_K) + \aleph.$$

c and h are divided by ξ and v is not, because the child consumes none of the drug: the child's bundle carries the Stone–Geary shifter $\bar{b}$ alone. The drug still reaches the child, through the *budget*. It is bought out of the same resources as consumption and rent, so it lowers the child's c and can also tip the household out of the rental and onto the street — and both c and h are in X_K. The child's weight is $\xi - 1$, so the household's weights sum to ξ, the same ξ that deflates its bundle. $\aleph$ is added once per household, not once per member.

Onset and mortality. Onset is a *flat* rate, not a function of how much is bought: the disorder arrives at the transition with probability π if $v > 0$ and with probability zero if $v = 0$. Survival is s_0 when clean and $s_1 < s_0$ with the disorder. The extensive margin carries all of the risk and the intensive margin none.

The problem. Write $\mathbf{1}_v = \mathbf{1}\{v > 0\}$, so the onset probability is $\pi \mathbf{1}_v$. The young household chooses (v, h, a') to maximise

$$\boxed{W_1(y) = u(X_P) + \omega_c(\xi - 1) u(X_K) + \aleph + \beta\Big[(1 - \pi\mathbf{1}_v)\, s_0 W_2(a', y, 0) + \pi\mathbf{1}_v\, s_1 W_2(a', y, 1)\Big] + \beta\, \omega_c\, \mathbb{E}\big[W_1(y_c)\big]}$$

The last term is the child's own value as a young adult, at its inherited income y_c, which depends on the disorder d' the parent ends with:

$$\mathbb{E}\big[W_1(y_c)\big] = (1 - \pi\mathbf{1}_v)\, W_1(y_c \,|\, d' = 0) + \pi\mathbf{1}_v\, W_1(y_c \,|\, d' = 1) \;=\; W_1(y_c \,|\, 0) - \pi\mathbf{1}_v\, \Delta(y),$$

$$\Delta(y) \;\equiv\; W_1(y_c \,|\, d' = 0) - W_1(y_c \,|\, d' = 1).$$

Only the second piece moves with the household's choice, so the drug decision faces exactly $-\beta\omega_c\pi\mathbf{1}_v\Delta(y)$: the child's loss, weighted by how likely the parent is to acquire the disorder at all. W_1 is on both sides of its own definition, which is what makes it a fixed point.

Survival sits *inside* the expectation over the realised disorder, not outside it: written as $\phi\,\mathbb{E}W_2$, the disorder becomes something the household wants wherever $W_2 < 0$. The $v = 0$ branch is solved on its own, with the clean continuation and the voucher unrevoked, and the household uses if the best $v > 0$ beats it.

Period 2 — terminal

Employment is redrawn, at a worse rate with the disorder, and the voucher is redrawn subject to the policy rule. Assets are run down and nothing is bequeathed, so with cash $= Ra +$ inc and $\xi = 1$,

$$X_2 = c^{\varphi_c}\, h^{\varphi_h}\, \bar{b}^{\,\varphi_q}, \qquad W_2(a, y, d) = \max_{h \in \{h_1, h_2\}} u(X_2) + \aleph, \qquad c = \text{cash} - \text{rent}(h, r).$$

The same $u(x) = x^{1-\sigma}/(1-\sigma)$ over the same Cobb–Douglas bundle as period 1, with only two differences, both because the child has left and there is no drug: $\xi = 1$, so nothing is deflated, and the drug shifter enters at $\bar{b}$ rather than $\bar{b} + v$. No drug choice, no child term, no recovery.

Policy. Housing First pays the voucher to every eligible household. Treatment First withdraws it from any household failing the sobriety condition *in the period the voucher is paid* — current use, or a current disorder. That single difference is the whole experiment.

The child

Income is inherited with persistence ρ, with the innovation that keeps the cross-section stationary:

$$\log y_c = \rho \log y + (1-\rho) \log \bar{y} - \delta_d \mathbf{1}\{d' = 1\} + \varepsilon, \qquad \varepsilon \sim N\big(0, (1-\rho^2)\sigma_y^2\big).$$

δ_d is the only inherited penalty: persistence alone delivers the homelessness odds ratio, so there is no separate homelessness penalty and the mixture has two types. The child is then run through the model as an adult at its own income, which is what gives it a disorder probability at all.

Both branches of $\Delta(y)$ are expectations of W_1 over the child's income draw, taken with the same kernel and differing only by whether δ_d is subtracted from its mean. The fixed point is a well behaved one: the $\aleph$ levels cancel in the difference. There is no survival factor on it — the child becomes young whether or not the parent lives, so s_d belongs on W_2 and on nothing else.

Lifetime utility

For welfare, W_1 is taken as the household's *own* value, with the internalised child term netted out, so that a planner counts the child once rather than once in full and again at ω_c. The childhood a cohort actually lives, at its parent's realised allocation, is

$$W_0 = (\xi - 1)\big[u(X_K) + \aleph\big], \qquad \text{and lifetime utility is} \qquad W = W_0 + \beta W_1,$$

with childhood lived under the policy one period earlier and adulthood under the policy now — the asymmetry a static comparison sums away, since the cohort the reform lands on has its childhood already behind it.

Consumption equivalents are CRRA, not log. Scaling c by $(1+\lambda)$ in every state scales both bundles by $(1+\lambda)^{\varphi_c}$, so every u term scales by $(1+\lambda)^{\varphi_c(1-\sigma)}$ while the $\aleph$ terms do not move:

$$\lambda = \Big(1 + \tfrac{dW}{U}\Big)^{\frac{1}{\varphi_c(1-\sigma)}} - 1, \qquad U = W - A, \qquad A = \aleph\Big(1 + \beta\big[(1-\pi)s_0 + \pi s_1\big]\Big),$$

which depends on the *level* through U, as the log formula never did.

What identifies what

γ	housing share of non-drug spending	θ	drug spending share among users
$\aleph$	value of a statistical life	π_{base}	disorder rate
h_1 or $p_r h_2$	homelessness rate	ω_c	the child use gap
$\bar{b}$	drug participation	δ_d	the child's SUD odds ratio

σ, β, ρ, s_0, the unemployment rates, the unit sizes and the voucher rule are set rather than calibrated. Participation and the disorder rate are tied by sud $=$ participation $\times\, \pi_{\text{base}}$, so they cannot be hit independently. ω_c acts in two places, the flow and the continuation, and the continuation half is $\Delta(y)$.